DPA Device — Perfboard Soldering Guide

DPA (Digital Playback Asset) — brought to you by **The DPAC** (Digital Playback Asset Consortium)

Firmware: v1.0.0 | **Board:** 7×9cm double-sided perfboard | **MCU:** ESP32-WROVER-32 DevKit V1 (30-pin)

Required Board

7×9cm double-sided prototype perfboard (2.54mm pitch, plated through-holes)

- **Columns:** A through Z (26 usable positions, 7cm)
- **Rows:** 1 through 34 (34 usable positions, 9cm)
- Grid reference format: **Column + Row** (e.g., `G3`, `T8`, `P15`)
- Orientation: **Portrait** (7cm wide, 9cm tall)
- Row 1 = TOP edge, Row 34 = BOTTOM edge
- Column A = LEFT edge, Column Z = RIGHT edge

> ***Important:*** *The small 5×3.5cm board in your photo is too small. Get a* ***7×9cm*** *board — search "7x9cm prototype PCB double sided" on Amazon/AliExpress.*

Bill of Materials (Solder BOM)

#	Component	Qty	Spec	Placement	Notes
1	Female pin header (1×15)	2	2.54mm pitch, 15-pin straight	G3–G17, P3–P17	ESP32 DevKit sockets
2	Female pin header (1×7)	1	2.54mm pitch, 7-pin straight	T5–T11	PCM5122 DAC socket
3	Female pin header (1×6)	1	2.54mm pitch, 6-pin straight	B20–B25	XTSD flash card socket
4	Tactile switch 6×6mm	5	4-pin through-hole, momentary	See Button Map	PLAY, PAUSE, NEXT, PREV, MODE
5	100K ohm resistor	2	1/4W, axial through-hole	D7–D9, D9–D10	Battery voltage divider
6	100nF ceramic capacitor	2	MLCC or disc, 2.54mm lead pitch	D10–E10, T12–U12	ADC filter + DAC decoupling
7	10uF electrolytic capacitor	1	16V, radial, 2mm lead spacing	U13–U14	DAC power decoupling
8	330 ohm resistor	1	1/4W, axial through-hole	V17–V19	LED data line protection
9	SN74LVC1T45 breakout or single FET level shifter	1	SOT-23-5 breakout or 3-pin module	V15–V19 area	3.3V→5V for WS2812B (optional)
10	2-pin screw terminal	3	5mm pitch, PCB mount	See Connector Map	Battery, LED strip power, LED strip data
11	3.5mm stereo audio jack	1	PJ-307 or similar, 5-pin PCB mount	S28–V30	Line-level audio output

12	4-pin JST-XH header	1	2.54mm pitch, right-angle or straight	U31–X31	PN532 NFC off-board cable
13	30 AWG wire-wrap wire	1 roll	Multiple colors (see wire table)	—	Point-to-point wiring on solder side
14	ESP32-WROVER-32 DevKit V1	1	30-pin (15 per side)	Plugs into G3–G17 / P3–P17	NOT soldered — sits in female headers
15	PCM5122 DAC breakout module	1	Generic 7-10 pin, I2S+I2C	Plugs into T5–T11	NOT soldered — sits in female header
16	Adafruit XTSD SPI Flash SD Card #6039 (4GB)	1	6-pin	Plugs into B20–B25	NOT soldered — sits in female header
17	PN532 NFC module	1	I2C mode, 4-wire cable	OFF-BOARD via JST cable	Mounted separately in enclosure
18	4.7K ohm resistor	2	1/4W, axial	R6–R3, S9–S3	I2C pull-ups (SDA + SCL) to 3.3V
19	1000uF electrolytic capacitor	1	10V+, radial	OFF-BOARD at LED strip	Placed at LED strip power input

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ESP32 DevKit V1 (30-Pin) Pinout Reference

This is the pin order when the DevKit is plugged in with USB port facing Row 1 (top edge).

Left Header — Column G (Rows 3–17)

Grid	Row	ESP32 Pin	GPIO	Function in DPA
G3	3	3V3	—	3.3V power rail
G4	4	EN	—	Enable (leave unconnected)
G5	5	VP	GPIO 36	— (unused, input only)
G6	6	VN	GPIO 39	— (unused, input only)
G7	7	**D34**	**GPIO 34**	**Battery ADC input**
G8	8	D35	GPIO 35	— (unused, input only)
G9	9	**D32**	**GPIO 32**	**Button: PREV**
G10	10	**D33**	**GPIO 33**	**Button: MODE**
G11	11	**D25**	**GPIO 25**	**Button: PLAY**
G12	12	**D26**	**GPIO 26**	**Button: PAUSE**
G13	13	**D27**	**GPIO 27**	**Button: NEXT**
G14	14	**D14**	**GPIO 14**	**XTSD SCK (HSPI)**
G15	15	**D12**	**GPIO 12**	**XTSD MISO (HSPI)**
G16	16	GND	—	Ground rail

G17	17	**D13**	**GPIO 13**	**XTSD MOSI (HSPI)**

Right Header — Column P (Rows 3–17)

Grid	Row	ESP32 Pin	GPIO	Function in DPA
P3	3	VIN	—	5V input (from USB or battery)
P4	4	GND	—	Ground rail
P5	5	**D23**	**GPIO 23**	**I2S DOUT → PCM5122 DIN**
P6	6	**D22**	**GPIO 22**	**I2C SCL → PCM5122 + PN532**
P7	7	TX	GPIO 1	Serial TX (leave for debug)
P8	8	RX	GPIO 3	Serial RX (leave for debug)
P9	9	**D21**	**GPIO 21**	**I2C SDA → PCM5122 + PN532**
P10	10	**D19**	**GPIO 19**	**I2S LRCLK → PCM5122 LCK**
P11	11	D18	GPIO 18	— (unused, reserve)
P12	12	**D5**	**GPIO 5**	**LED Strip Data (WS2812B)**
P13	13	D17	GPIO 17	**DO NOT USE** (WROVER PSRAM)
P14	14	D16	GPIO 16	**DO NOT USE** (WROVER PSRAM)
P15	15	**D4**	**GPIO 4**	**I2S BCLK → PCM5122 BCK**
P16	16	D2	GPIO 2	Onboard LED (optional)
P17	17	**D15**	**GPIO 15**	**XTSD CS (HSPI)**

Board Layout Map

```
COMPONENT SIDE (TOP VIEW) — 7×9cm Perfboard
USB port faces up (Row 1)

     A  B  C  D  E  F  G  H  I  J  K  L  M  N  O  P  Q  R  S  T  U  V  W  X  Y  Z
     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
 1 ──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○── TOP EDGE
 2 ──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──
 3 ──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──○──○── G3=3V3        P3=VIN
 4 ──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──○──○── G4=EN         P4=GND
 5 ──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──○──■──○──○──○──▌──○──○──○──○──○──○── G5=VP
P5=GPIO23    T5=PCM VCC
 6 ──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──○──■──○──●──○──▌──○──○──○──○──○──○── G6=VN
P6=GPIO22    T6=PCM GND  R6=I2C pullup
 7 ──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──○──■──○──○──○──▌──○──○──○──○──○──○── G7=GPIO34     P7=TX
T7=PCM BCK
 8 ──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──○──■──○──○──○──▌──○──○──○──○──○──○── G8=GPIO35     P8=RX
T8=PCM DIN
 9 ──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──○──■──○──●──○──▌──○──○──○──○──○──○── G9=GPIO32
```

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P9=GPIO21    T9=PCM LCK  R9=I2C pullup
10 ──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──■──○──○──○──▐──○──○──○──○──○──○──  G10=GPIO33
P10=GPIO19   T10=PCM SCL
11 ──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──■──○──○──○──▐──○──○──○──○──○──○──  G11=GPIO25
P11=GPIO18   T11=PCM SDA
12 ──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──○──○──  G12=GPIO26
P12=GPIO5
13 ──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──○──○──  G13=GPIO27
P13=GPIO17 ⛔
14 ──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──○──○──  G14=GPIO14
P14=GPIO16 ⛔
15 ──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──○──○──  G15=GPIO12
P15=GPIO4
16 ──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──○──○──  G16=GND
P16=GPIO2
17 ──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──■──○──○──○──○──○──○──○──○──○──○──  G17=GPIO13
P17=GPIO15
     │  │  │  │  │  │  │  │  │  │  │  │  │  │  │  │  │  │  │  │  │  │  │  │  │  │  │  │
     │  │  │  │  │  │  └──── ESP32 DevKit ──────┘  │  │  │  │  │  │  │  │  │  │  │
18 ──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──
     │  │  │  │  │  │                                                │  │  │  │
19 ──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──
     │  │                                                            │  │  │  │
20 ──○──▐──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──  B20=XTSD VIN
21 ──○──▐──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──  B21=XTSD GND
22 ──○──▐──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──  B22=XTSD SCK
23 ──○──▐──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──  B23=XTSD MISO
24 ──○──▐──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──  B24=XTSD MOSI
25 ──○──▐──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──  B25=XTSD CS
     │  │                                                            │  │  │  │
26 ──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──
     │     │     │     │     │     │     │     │     │     │     │     │     │
27 ──○──○──○──○──○──○──○──○──○──BTN──BTN──BTN──BTN──BTN──○──○──○──○──○──○──○──○──○──○──○──
28 ──○──○──○──○──○──○──○──○──○──PLAY─PAUSE NEXT─PREV─MODE──○──○──○──○──○──○──○──○──○──○──
29 ──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──
     │                                                                  │  │  │  │  │
30 ──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──■──■──■──■──○──○──○──  T30─W30: Audio Jack
31 ──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──■──■──■──■──○──○──  U31─X31: NFC JST
header
32 ──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──
33 ──[BATT+ ][BATT- ]──○──○──[LED 5V ][LED GND][LED DIN]──○──○──○──○──○──○──○──○──○──  Screw terminals
34 ──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──○──  BOTTOM EDGE

LEGEND:
  ■ = Female header pin (solder socket from bottom, module plugs in from top)
  ● = I2C pull-up resistor pad
  ⛔ = Do NOT use (WROVER PSRAM conflict)
  ○ = Open hole (available for wiring)
```

Component Placement — Pin by Pin

1. ESP32 DevKit V1 Female Header Sockets

Solder **two 1×15 female pin headers** from the bottom side of the board. The ESP32 DevKit will plug in from the top.

Header	Column	Start Row	End Row	Notes
Left socket	**G**	**3**	**17**	15 pins, solder from bottom
Right socket	**P**	**3**	**17**	15 pins, solder from bottom

Verification: The distance from column G to column P = 9 holes = 22.86mm. This matches the ESP32 DevKit V1 (30-pin) row spacing. If your DevKit has 10-hole spacing (25.4mm), use columns **G** and **Q** instead.

2. PCM5122 DAC Module Female Header Socket

Solder **one 1×7 female pin header** in Column T. Your PCM5122 breakout module plugs in from the top.

Pin #	Grid	Module Pin	Description
1	**T5**	VCC	3.3V power
2	**T6**	GND	Ground
3	**T7**	BCK	Bit clock (I2S BCLK)
4	**T8**	DIN	Data in (I2S audio data)
5	**T9**	LCK	Left/right clock (I2S LRCLK)
6	**T10**	SCL	I2C clock
7	**T11**	SDA	I2C data

> ***Note:*** *PCM5122 breakout pin orders vary by manufacturer. Check YOUR module's silkscreen and match the pin NAMES (VCC, GND, BCK, DIN, LCK, SCL, SDA) to the correct socket positions. Re-order if needed.*

If your module has extra pins (FMT, XSMT, DEMP, FLT), extend the header to T12, T13, etc., and:

- **FMT** → tie to GND (I2S mode)
- **XSMT** → tie to 3.3V (unmuted)
- **DEMP** / **FLT** → leave unconnected

3. XTSD SPI Flash SD Card (Adafruit #6039) Female Header Socket

Solder **one 1×6 female pin header** in Column B. The XTSD is a 4GB soldered flash chip that emulates an SD card — same SPI pins, same `SD.h` library, no firmware changes.

Pin #	Grid	Module Pin	Description
1	**B20**	VIN	3.3V power (onboard regulator accepts 3V–5V)
2	**B21**	GND	Ground
3	**B22**	SCK	SPI clock
4	**B23**	MISO	Master In / Slave Out (data out from flash)
5	**B24**	MOSI	Master Out / Slave In (data in to flash)
6	**B25**	CS	Chip select (has onboard pull-up)

4. Tactile Buttons (5×)

Standard 6mm tactile buttons need their legs bent slightly to fit the 2.54mm grid. Each button uses **2 active pins** (the other 2 are internally shorted duplicates).

Mount all 5 buttons in a row across Row 27–28:

Button	Function	Pin A (GPIO) Grid	Pin B (GND) Grid
BTN1	**PLAY**	**H27**	**H29**
BTN2	**PAUSE**	**J27**	**J29**
BTN3	**NEXT**	**L27**	**L29**
BTN4	**PREV**	**N27**	**N29**
BTN5	**MODE**	**P27**	**P29**

Orientation: Each button bridges 2 rows (27 and 29). Pin A (top, row 27) connects to its GPIO via wire. Pin B (bottom, row 29) connects to GND bus.

GND bus for buttons: Run a bare wire along **Row 29 from H29 to P29**, soldered to each button's bottom pin. Then connect one end to a GND point.

5. I2C Pull-Up Resistors (2× 4.7K ohm)

Resistor	From	To	Description
R_SCL	**R6**	**R3**	SCL pull-up: R6 is near GPIO 22 (SCL), R3 bridges to 3.3V rail at Row 3
R_SDA	**S9**	**S3**	SDA pull-up: S9 is near GPIO 21 (SDA), S3 bridges to 3.3V rail at Row 3

Wire **R3** to **G3** (3.3V) and **S3** to **G3** (3.3V) to complete the pull-up path.

6. Battery Voltage Divider (2× 100K ohm)

```
            R1 (100K)         R2 (100K)
BATT+ ────[D7]────[D9]────[D10]──── GND
                    │
                    └── GPIO 34 (at G7, connected via wire from D9)
```

Component	From Grid	To Grid	Description
R1 (100K top)	**D7**	**D9**	Upper divider resistor
R2 (100K bottom)	**D9**	**D11**	Lower divider resistor
C_filter (100nF)	**E9**	**E11**	Noise filter cap, parallel with R2

- Wire **D7** → screw terminal BATT+ at **A33**
- Wire **D9** → **G7** (GPIO 34 ADC input) — short jumper wire
- Wire **D11** → GND bus

7. Decoupling Capacitors

Capacitor	(+) Grid	(−) Grid	For	Description
100nF ceramic	**T12**	**U12**	PCM5122	Place right below DAC header, bridge T12 to GND

10uF electrolytic	**T13** (+)	**U13** (–)	PCM5122	Bulk decoupling, long leg (+) in T13

Wire **U12** and **U13** to the nearest GND point (P4 via a wire).

8. LED Data Line Resistor (330 ohm)

From Grid	To Grid	Description
V16	**V18**	330 ohm in series on LED data line

Wire **P12** (GPIO 5) → **V16** (resistor input). Wire **V18** (resistor output) → LED DIN screw terminal at Row 33.

9. 3.5mm Audio Jack (PJ-307 or similar)

Mount at the **bottom-right area** of the board:

Jack Pin	Grid	Connection
Tip (Left audio)	**T30**	Wire to PCM5122 VOUTL (off-board from module)
Ring (Right audio)	**U30**	Wire to PCM5122 VOUTR (off-board from module)
Sleeve (Ground)	**V30**	Wire to audio GND (PCM5122 AGND)

> ***Note:*** *The PCM5122 module's analog outputs (VOUTL, VOUTR, AGND) connect to the audio jack via short wires from the module pins. These are NOT routed through the female header — they come directly off the module's output pins.*

10. NFC Off-Board Connector (4-pin JST-XH)

Solder a 4-pin JST-XH header at the bottom-right:

Pin #	Grid	Signal	Cable Color
1	**U31**	3.3V	Red
2	**V31**	GND	Black
3	**W31**	SDA	Blue
4	**X31**	SCL	Yellow

The PN532 NFC module mounts **separately in the enclosure** and connects via a 4-wire cable to this header.

11. Screw Terminals (Bottom Edge, Row 33)

Terminal	Grid	Signal	Wire Color
BATT+	**A33**	Battery positive (3.0–4.2V)	Red
BATT–	**B33**	Battery negative / GND	Black
LED 5V	**G33**	5V power to LED strip	Red
LED GND	**H33**	LED strip ground	Black
LED DIN	**I33**	LED strip data in	Green

Master Wire Table

All wires run on the **SOLDER SIDE (bottom)** of the board. Use **30 AWG wire-wrap wire** in the color indicated.

I2S Audio Bus (ESP32 → PCM5122 DAC)

Wire #	From	Signal	To	Color	Length
W1	**P15** (GPIO 4)	I2S BCLK	**T7** (PCM BCK)	Orange	~3cm
W2	**P5** (GPIO 23)	I2S DOUT	**T8** (PCM DIN)	Yellow	~3cm
W3	**P10** (GPIO 19)	I2S LRCLK	**T9** (PCM LCK)	Green	~3cm

I2C Bus (ESP32 → PCM5122 + NFC)

Wire #	From	Signal	To	Color	Length
W4	**P6** (GPIO 22)	I2C SCL	**T10** (PCM SCL)	Blue	~3cm
W5	**P6** (GPIO 22)	I2C SCL	**X31** (NFC SCL)	Blue	~8cm
W6	**P9** (GPIO 21)	I2C SDA	**T11** (PCM SDA)	White	~3cm
W7	**P9** (GPIO 21)	I2C SDA	**W31** (NFC SDA)	White	~7cm

> ***I2C Note:*** *W4 and W5 share the same source (P6). Solder both wires to pad P6. Same for W6 and W7 sharing P9.*

I2C Pull-Up Resistors

Wire #	From	Signal	To	Color	Length
W8	**R6** (top of R_SCL) → resistor → **R3**	SCL pull-up	—	—	Resistor body
W9	**R3**	3.3V for SCL pull-up	**G3** (3V3)	Red	~3cm
W10	**S9** (top of R_SDA) → resistor → **S3**	SDA pull-up	—	—	Resistor body
W11	**S3**	3.3V for SDA pull-up	**G3** (3V3)	Red	~3cm
W12	**R6**	SCL tap	**P6** (GPIO 22)	Blue	~1cm
W13	**S9**	SDA tap	**P9** (GPIO 21)	White	~1cm

HSPI SD Card Bus (ESP32 → XTSD Flash)

Wire #	From	Signal	To	Color	Length
W14	**G14** (GPIO 14)	XTSD SCK	**B22** (XTSD SCK)	Purple	~4cm
W15	**G15** (GPIO 12)	XTSD MISO	**B23** (XTSD MISO)	Brown	~4cm
W16	**G17** (GPIO 13)	XTSD MOSI	**B24** (XTSD MOSI)	Gray	~4cm
W17	**P17** (GPIO 15)	XTSD CS	**B25** (XTSD CS)	Tan	~6cm

> ***W17 Note:*** *This wire crosses the full width of the board (column P → column B). Route it along the bottom edge to keep it tidy.*

LED Strip Data (ESP32 → Level Shifter → LED)

Wire #	From	Signal	To	Color	Length
W18	**P12** (GPIO 5)	LED data out	**V16** (330Ω resistor in)	Green	~2cm
W19	**V18** (330Ω resistor out)	LED data (protected)	**I33** (LED DIN terminal)	Green	~5cm

Battery ADC

Wire #	From	Signal	To	Color	Length
W20	**D9** (divider midpoint)	ADC voltage	**G7** (GPIO 34)	Orange	~1cm
W21	**D7** (divider top)	BATT+	**A33** (BATT+ terminal)	Red	~5cm

Button Wiring

Wire #	From	Signal	To	Color	Length
W22	**H27** (BTN1 top)	PLAY	**G11** (GPIO 25)	Blue	~5cm
W23	**J27** (BTN2 top)	PAUSE	**G12** (GPIO 26)	Blue	~5cm
W24	**L27** (BTN3 top)	NEXT	**G13** (GPIO 27)	Blue	~5cm
W25	**N27** (BTN4 top)	PREV	**G9** (GPIO 32)	Blue	~5cm
W26	**P27** (BTN5 top)	MODE	**G10** (GPIO 33)	Blue	~5cm

Power Distribution

Wire #	From	Signal	To	Color	Length
W27	**G3** (3V3)	3.3V power	**T5** (PCM VCC)	Red	~4cm
W28	**G3** (3V3)	3.3V power	**B20** (XTSD VIN)	Red	~5cm
W29	**G3** (3V3)	3.3V power	**U31** (NFC 3.3V)	Red	~6cm
W30	**G16** (GND)	Ground	**T6** (PCM GND)	Black	~4cm
W31	**G16** (GND)	Ground	**B21** (XTSD GND)	Black	~5cm
W32	**G16** (GND)	Ground	**V31** (NFC GND)	Black	~6cm
W33	**G16** (GND)	Ground	**B33** (BATT– terminal)	Black	~5cm
W34	**G16** (GND)	Ground	**H33** (LED GND terminal)	Black	~5cm
W35	**P4** (GND)	Ground	**U12** (DAC decoupling cap –)	Black	~2cm
W36	**P4** (GND)	Ground	**D11** (voltage divider bottom)	Black	~4cm
W37	**P4** (GND)	Ground	**Row 29 bus** (button GND)	Black	~4cm
W38	**P3** (VIN 5V)	5V power	**G33** (LED 5V terminal)	Red	~5cm

Button GND Bus

Wire #	From	To	Color	Description
W39	**H29**	**P29**	Black	Bare wire along Row 29, soldered at H29, J29, L29, N29, P29
W40	**H29**	**P4** (or G16)	Black	GND bus to ground rail

Total Wire Count Summary

Category	Wires	Color Code
I2S Audio (3 wires)	W1–W3	Orange, Yellow, Green
I2C Bus (4 wires + 2 taps)	W4–W7, W12–W13	Blue, White
I2C Pull-ups (4 wires + 2 resistors)	W8–W11	Red (power wires)
HSPI XTSD Flash (4 wires)	W14–W17	Purple, Brown, Gray, Tan
LED Strip (2 wires + resistor)	W18–W19	Green
Battery ADC (2 wires + 2 resistors + cap)	W20–W21	Orange, Red
Buttons (5 wires + bus)	W22–W26, W39–W40	Blue, Black
Power rails (12 wires)	W27–W38	Red (power), Black (ground)
TOTAL	**40 wires + 7 passive components**	

Soldering Order (Step by Step)

Follow this exact order. Each step must be completed before the next.

Step	Action	Components	Verify
1	Solder female headers for ESP32	2× 1×15 at G3–G17 and P3–P17	Headers straight, pins accessible from top
2	Solder female header for PCM5122	1× 1×7 at T5–T11	Header straight
3	Solder female header for XTSD flash	1× 1×6 at B20–B25	Header straight
4	Solder I2C pull-up resistors	4.7K at R3–R6 and S3–S9	Resistor values correct (yellow-violet-red)
5	Solder voltage divider resistors	2× 100K at D7–D9 and D9–D11	Resistor values correct (brown-black-yellow)
6	Solder filter cap (100nF)	C_filter at E9–E11	No polarity (ceramic)
7	Solder DAC decoupling caps	100nF at T12–U12, 10uF at T13–U13	Electrolytic: long leg (+) in T13
8	Solder 330Ω LED resistor	At V16–V18	(orange-orange-brown)
9	Solder screw terminals	Row 33: A33/B33, G33/H33, I33	Openings face board edge

10	Solder NFC JST header	U31–X31 (4-pin)	Opening faces board edge
11	Solder audio jack	T30–V30 area	Test fit first
12	Solder tactile buttons	H27, J27, L27, N27, P27 (5 buttons)	All buttons click when pressed
13	**FLIP BOARD — solder side up**		
14	Solder button GND bus wire	W39: bare wire H29 → P29	Solder at each button's GND pin
15	Solder I2S wires (W1–W3)	P15→T7, P5→T8, P10→T9	Orange, Yellow, Green
16	Solder I2C wires (W4–W7, W12–W13)	P6→T10, P6→X31, P9→T11, P9→W31	Blue, White
17	Solder I2C pull-up power wires	W9: R3→G3, W11: S3→G3	Red
18	Solder HSPI SD wires (W14–W17)	G14→B22, G15→B23, G17→B24, P17→B25	Purple, Brown, Gray, Tan
19	Solder LED data wire (W18–W19)	P12→V16, V18→I33	Green
20	Solder battery ADC wire (W20–W21)	D9→G7, D7→A33	Orange, Red
21	Solder button signal wires (W22–W26)	H27→G11, J27→G12, L27→G13, N27→G9, P27→G10	All Blue
22	Solder power wires (W27–W29)	G3→T5, G3→B20, G3→U31	All Red
23	Solder ground wires (W30–W37)	G16/P4 to T6, B21, V31, B33, H33, U12, D11, Row29	All Black
24	Solder 5V LED power wire (W38)	P3→G33	Red
25	**VISUAL INSPECTION**		No solder bridges, no cold joints
26	Plug in ESP32 DevKit	Into G3–G17 / P3–P17 headers	USB port faces Row 1 (top)
27	Plug in PCM5122 module	Into T5–T11 header	Check pin order matches
28	Plug in XTSD flash card	Into B20–B25 header	Check pin order matches

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Verification Checklist (Before Power On)

Use a multimeter in continuity mode. Test each connection from the table below.

Test #	Probe A	Probe B	Expected	What It Checks
V1	G3 (3V3)	T5 (PCM VCC)	Continuity	DAC power
V2	G3 (3V3)	B20 (XTSD VIN)	Continuity	XTSD power
V3	G3 (3V3)	U31 (NFC 3V3)	Continuity	NFC power
V4	G16 (GND)	T6 (PCM GND)	Continuity	DAC ground

V5	G16 (GND)	B21 (XTSD GND)	Continuity	XTSD ground
V6	G16 (GND)	V31 (NFC GND)	Continuity	NFC ground
V7	G16 (GND)	B33 (BATT–)	Continuity	Battery ground
V8	G3 (3V3)	G16 (GND)	**NO continuity**	**No short on power rail**
V9	P3 (VIN)	P4 (GND)	**NO continuity**	**No short on 5V rail**
V10	P15 (GPIO 4)	T7 (PCM BCK)	Continuity	I2S BCLK wire
V11	P5 (GPIO 23)	T8 (PCM DIN)	Continuity	I2S DOUT wire
V12	P10 (GPIO 19)	T9 (PCM LCK)	Continuity	I2S LRCLK wire
V13	P6 (GPIO 22)	T10 (PCM SCL)	Continuity	I2C SCL wire
V14	P9 (GPIO 21)	T11 (PCM SDA)	Continuity	I2C SDA wire
V15	G14 (GPIO 14)	B22 (XTSD SCK)	Continuity	HSPI CLK wire
V16	G15 (GPIO 12)	B23 (XTSD MISO)	Continuity	HSPI MISO wire
V17	G17 (GPIO 13)	B24 (XTSD MOSI)	Continuity	HSPI MOSI wire
V18	P17 (GPIO 15)	B25 (XTSD CS)	Continuity	HSPI CS wire
V19	Press each button	G11/G12/G13/G9/G10 to G16	Continuity when pressed	Button wiring
V20	G7 (GPIO 34)	D9 (divider mid)	Continuity	Battery ADC wire

> ***CRITICAL:*** *Tests V8 and V9 must show NO continuity. If either shows continuity, there is a short circuit. Find and fix before applying power.*

Off-Board Connections Reference

Component	Cable	Connects To On Board	Notes
WS2812B LED strip	3-wire (5V, GND, DIN)	G33 (5V), H33 (GND), I33 (DIN)	Add 1000uF cap across 5V/GND at strip end
LiPo Battery	2-wire (red +, black –)	A33 (BATT+), B33 (BATT–)	Through TP4056 charger module
PN532 NFC module	4-wire JST cable	U31 (3V3), V31 (GND), W31 (SDA), X31 (SCL)	Set PN532 DIP: SEL0=ON, SEL1=OFF
3.5mm AUX out	Wired from PCM5122 module	T30 (L), U30 (R), V30 (GND)	Via 470uF coupling caps recommended
USB-C	Built into ESP32 DevKit	Plugged into board headers	For programming + charging